

DANDI:000458: Simultaneous electroencephalography, extracellular electrophysiology, and cortical electrical stimulation in head-fixed mice

6 of the 24 NWB files contain no units table rows (see section 3). Region labels are the units.location values stored in the files.

Scope of this export (computed by this export): the units, electrodes and intervals tables of every NWB file were read remotely from the DANDI Archive; voltage, LFP, video and other time series were not downloaded.

Source labels: **DANDI metadata** = the Dandiset's published metadata; **NWB file** = values or text stored in the NWB files; **Computed by this export** = derived by the export scripts in `dandi_export/` (definitions given where used).

1. The dataset as published

Everything in this section is quoted from the Dandiset's published metadata or, for file-level fields, from `sub-521885_ses-20200709_behavior+ecephys.nwb`.

Field	Value (DANDI metadata)
Name	Simultaneous electroencephalography, extracellular electrophysiology, and cortical electrical stimulation in head-fixed mice
Identifier / version	DANDI:000458 / 0.230317.0039
Citation	Claar, Leslie D; Rembado, Irene; Kuyat, Jacquelyn R; Russo, Simone; Marks, Lydia C; Olsen, Shawn R; Koch, Christof (2023) Simultaneous electroencephalography, extracellular electrophysiology, and cortical electrical stimulation in head-fixed mice (Version 0.230317.0039) [Data set]. DANDI archive. https://doi.org/10.48324/dandi.000458/0.230317.0039
DOI	10.48324/dandi.000458/0.230317.0039
Contributors (roles)	Claar, Leslie D (ContactPerson, Author); Rembado, Irene (Author, ContactPerson); Kuyat, Jacquelyn R (Author); Russo, Simone (Author); Marks, Lydia C (Author); Olsen, Shawn R (Author); Koch, Christof (Author); Tiny Blue Dot Foundation (Funder); Allen Institute (Sponsor, Affiliation)
License	CC-BY-4.0
Keywords	EEG, Neuropixels, electrical stimulation, brain states, cortico-thalamic interactions, extracellular electrophysiology
Related resources	Cortico-thalamo-cortical interactions modulate electrically evoked EEG responses in mice Reviewed preprint (eLife) (IsDescribedBy); Cortico-thalamo-cortical interactions modulate electrically evoked EEG responses in mice Manuscript preprint (bioRxiv) (IsDescribedBy)
Files / size / subjects	24 files, 361.48 GB, 23 subjects
Species	Mus musculus - House mouse
Approach	electrophysiological approach; behavioral approach
Measurement technique	surgical technique; spike sorting technique; behavioral technique; analytical technique; signal filtering technique; multi electrode extracellular electrophysiology recording technique
Variables measured	BehavioralTimeSeries, Units, ElectricalSeries, ElectrodeGroup, LFP, ProcessingModule
Date published	2023-03-17T00:39:49.682458+00:00

Description (DANDI metadata, verbatim)

In this set of experiments, we recorded neural signals using an electroencephalography (EEG) array and Neuropixels probes while directly electrically stimulating the cortex in awake and anesthetized head-fixed mice. Details of the experiments can be found in the associated manuscript (Claar, Rembado et al., 2023). Briefly, the recordings and stimulation commenced while the mouse was awake, free to rest or move on a rotating disc. After delivering a series of stimuli, anesthesia was induced with isoflurane (5% via inhalation). Once a surgical level of anesthesia was reached, the mouse was maintained unconscious (1-1.5% isoflurane) during the delivery of another series of stimuli. In some subjects another series of stimuli was delivered after the isoflurane was turned off, this epoch/state is referred to as "recovery" in the dataset.

Each session file includes the following data: raw EEG signals from all 30 surface electrodes and the associated electrode information; raw LFP signals from all electrodes on all Neuropixels probes (if used) and the associated electrode information; spike times for all units (spike sorted with Kilosort 2.0) that passed our quality threshold with associated information (brain region, spike waveform duration, etc.); subject's speed computed from the rotation of the disc for the entire session; epochs marking when isoflurane was induced and when it was maintained; and a trial table containing

information about every stimulus delivered throughout the session.

All sessions included an awake and an isoflurane epoch, but not all sessions included a recovery epoch. Some subjects received visual stimulation trials (visual stimuli presented on a screen in the subject's right field of view) interleaved with the electrical stimulation trials.

File-level fields (NWB file, verbatim)

Field	Value (NWB file)
session_description	EEG recording during wakefulness and isoflurane anesthesia
experiment_description	in vivo electrophysiology in a head-fixed mouse during cortical electrical microstimulation
institution	Allen Institute
experimenter	Claar, Leslie D; Rembado, Irene; Kuyat, Jacquelyn R; Russo, Simone; Marks, Lydia C; Olsen, Shawn R; Koch, Christof
keywords	EEG; Neuropixels; electrical stimulation; brain states; cortico-thalamic interactions
related_publications	https://doi.org/10.7554/eLife.84630.1
stimulus_notes	single pulse electrical stimuli targeted to MOs (superficial layers)
subject	subject_id: 521885; species: Mus musculus; strain: C57BL/6J; sex: M; age: P156D; description: mouse521885
devices	EEG array: H32 Mouse EEG (30-ch)
electrode_groups	EEG array: description = '30-ch surface grid', location = 'skull surface, both hemispheres'

2. Contents of the files

Quantity	Value	Source
NWB files read / subjects	24 / 23	NWB file
Species	Mus musculus (24)	NWB file
Sex (files)	M (14), F (10)	NWB file
Genotype (files)		NWB file
Strain (files)	C57BL/6J (24)	NWB file
Age (files)	P177D (2), P72D (2), P114D (2), P135D (2), P128D (2), P156D (1), P163D (1), P65D (1), P108D (1), P191D (1), P121D (1), P86D (1), P94D (1), P129D (1), P80D (1), P93D (1), P119D (1), P170D (1), P198D (1)	NWB file
Lab / institution	/ Allen Institute (24)	NWB file
Units (total)	12,149	NWB file
Units per file	median 560 (range 0–995)	NWB file
Spikes (sum of spike_times_index lengths)		NWB file
How region was assigned to units	units.location	Computed by this export
Distinct region labels	71	NWB file

The meaning of each units column, including any quality label, is quoted from the NWB file in section 7.

3. NWB files

File (session_id)	Subject	Sex	Age	Size (GB)	Units	Electrodes	Electrode groups	Interval tables (rows)
sub-521885_ses-20200709_behavior+ecephys	521885	M	P156D	0.31	0	30	EEG array	epochs(2); trials(360)
sub-521886_ses-20200716_behavior+ecephys	521886	M	P163D	0.50	0	30	EEG array	epochs(2); trials(900)
sub-521887_ses-20200730_behavior+ecephys	521887	F	P177D	0.43	0	30	EEG array	epochs(2); trials(800)
sub-543393_ses-20200820_behavior+ecephys	543393	M	P65D	0.53	0	30	EEG array	epochs(2); trials(900)
sub-543394_ses-20200827_behavior+ecephys	543394	M	P72D	0.59	0	30	EEG array	epochs(2); trials(900)
sub-546655_ses-20201023_behavior+ecephys	546655	F	P108D	14.87	265	798	EEG array, probeB, probeC	epochs(2); trials(1459)
sub-551397_ses-20210211_behavior+ecephys	551397	M	P191D	15.20	607	798	EEG array, probeB, probeF	epochs(2); trials(1800)
sub-551399_ses-20210128_behavior+ecephys	551399	F	P177D	11.69	593	798	EEG array, probeB, probeF	epochs(2); trials(1440)
sub-569062_ses-20210218_behavior+ecephys	569062	F	P72D	24.83	902	1182	EEG array, probeB, probeD, probeF	epochs(2); trials(1440)
sub-569064_ses-20210408_behavior+ecephys	569064	F	P121D	25.89	481	1182	EEG array, probeB, probeC, probeF	epochs(2); trials(1920)
sub-569068_ses-20210304_behavior+ecephys	569068	M	P86D	27.29	338	1182	EEG array, probeB, probeD, probeF	epochs(2); trials(1800)
sub-569069_ses-20210312_behavior+ecephys	569069	M	P94D	22.47	520	1182	EEG array, probeB, probeC, probeF	epochs(2); trials(1800)
sub-569070_ses-20210401_behavior+ecephys	569070	M	P114D	1.12	0	30	EEG array	epochs(2); trials(1440)
sub-569072_ses-20210422_behavior+ecephys	569072	M	P135D	13.30	995	1182	EEG array, probeB, probeC, probeF	epochs(2); trials(960)
sub-569073_ses-20210415_behavior+ecephys	569073	M	P128D	26.73	631	1182	EEG array, probeB, probeC, probeF	epochs(2); trials(2040)
sub-569073_ses-20210416_behavior+ecephys	569073	M	P129D	13.20	705	1182	EEG array, probeB, probeC, probeF	epochs(2); trials(720)
sub-571619_ses-20210319_behavior+ecephys	571619	M	P80D	23.17	859	1182	EEG array, probeB, probeC, probeF	epochs(2); trials(1440)
sub-571620_ses-20210513_behavior+ecephys	571620	F	P135D	23.25	771	1182	EEG array, probeB, probeC, probeF	epochs(2); trials(1680)
sub-575102_ses-20210603_behavior+ecephys	575102	F	P128D	24.33	528	1182	EEG array, probeB, probeC, probeF	epochs(2); trials(1882)
sub-586466_ses-20210729_behavior+ecephys	586466	M	P93D	14.28	470	798	EEG array, probeB, probeF	epochs(2); trials(1680)
sub-586468_ses-20210819_behavior+ecephys	586468	F	P114D	22.11	977	1182	EEG array, probeB, probeC, probeF	epochs(2); trials(1680)
sub-590479_ses-20210921_behavior+ecephys	590479	M	P119D	12.03	616	798	EEG array, probeB, probeF	epochs(2); trials(1440)
sub-590480_ses-20211111_behavior+ecephys	590480	F	P170D	21.09	948	1182	EEG array, probeB, probeC, probeF	epochs(2); trials(1680)

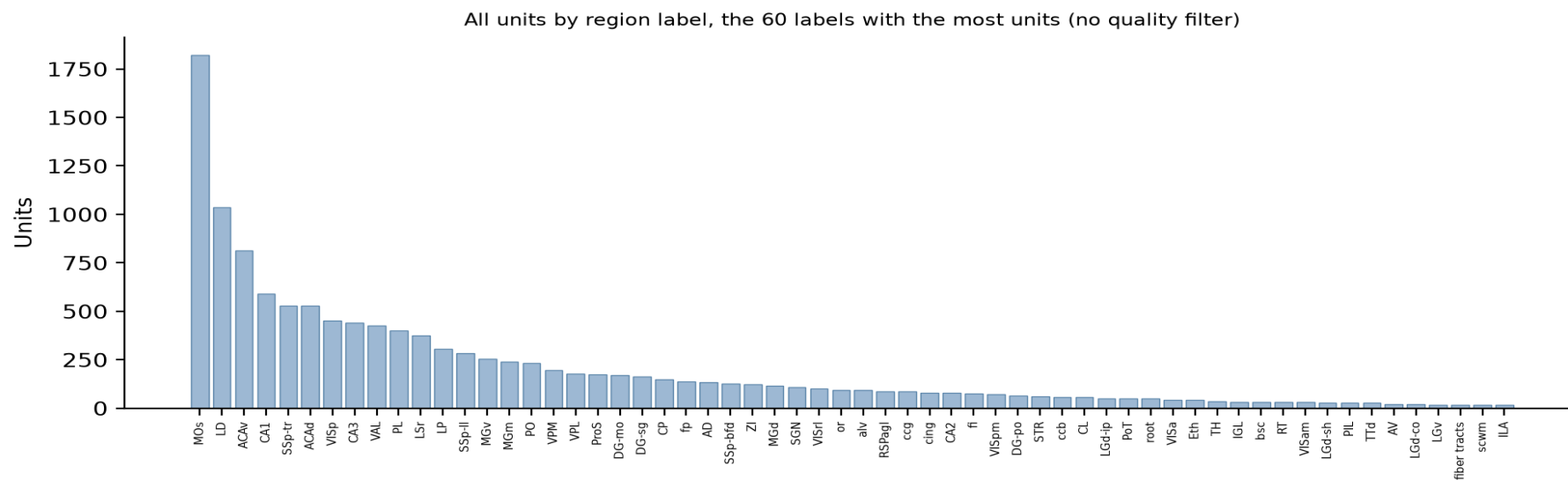
File (session_id)	Subject	Sex	Age	Size (GB)	Units	Electrodes	Electrode groups	Interval tables (rows)
sub-599017_ses-20220210_behavior+ecephys	599017	F	P198D	22.27	943	1182	EEG array, probeB, probeC, probeF	epochs(2); trials(1680)

Other data in each file (not exported)

File (session_id)	Acquisition	Processing modules: interfaces
sub-521885_ses-20200709_behavior+ecephys	ElectricalSeriesEEG	behavior:BehavioralTimeSeries
sub-521886_ses-20200716_behavior+ecephys	ElectricalSeriesEEG	behavior:BehavioralTimeSeries
sub-521887_ses-20200730_behavior+ecephys	ElectricalSeriesEEG	behavior:BehavioralTimeSeries
sub-543393_ses-20200820_behavior+ecephys	ElectricalSeriesEEG	behavior:BehavioralTimeSeries
sub-543394_ses-20200827_behavior+ecephys	ElectricalSeriesEEG	behavior:BehavioralTimeSeries
sub-546655_ses-20201023_behavior+ecephys	ElectricalSeriesEEG, LFPprobeB, LFPprobeC	behavior:BehavioralTimeSeries
sub-551397_ses-20210211_behavior+ecephys	ElectricalSeriesEEG, LFPprobeB, LFPprobeF	behavior:BehavioralTimeSeries
sub-551399_ses-20210128_behavior+ecephys	ElectricalSeriesEEG, LFPprobeB, LFPprobeF	behavior:BehavioralTimeSeries
sub-569062_ses-20210218_behavior+ecephys	ElectricalSeriesEEG, LFPprobeB, LFPprobeD, LFPprobeF	behavior:BehavioralTimeSeries
sub-569064_ses-20210408_behavior+ecephys	ElectricalSeriesEEG, LFPprobeB, LFPprobeC, LFPprobeF	behavior:BehavioralTimeSeries
sub-569068_ses-20210304_behavior+ecephys	ElectricalSeriesEEG, LFPprobeB, LFPprobeD, LFPprobeF	behavior:BehavioralTimeSeries
sub-569069_ses-20210312_behavior+ecephys	ElectricalSeriesEEG, LFPprobeB, LFPprobeC, LFPprobeF	behavior:BehavioralTimeSeries
sub-569070_ses-20210401_behavior+ecephys	ElectricalSeriesEEG	behavior:BehavioralTimeSeries
sub-569072_ses-20210422_behavior+ecephys	ElectricalSeriesEEG, LFPprobeB, LFPprobeC, LFPprobeF	behavior:BehavioralTimeSeries
sub-569073_ses-20210415_behavior+ecephys	ElectricalSeriesEEG, LFPprobeB, LFPprobeC, LFPprobeF	behavior:BehavioralTimeSeries
sub-569073_ses-20210416_behavior+ecephys	ElectricalSeriesEEG, LFPprobeB, LFPprobeC, LFPprobeF	behavior:BehavioralTimeSeries
sub-571619_ses-20210319_behavior+ecephys	ElectricalSeriesEEG, LFPprobeB, LFPprobeC, LFPprobeF	behavior:BehavioralTimeSeries
sub-571620_ses-20210513_behavior+ecephys	ElectricalSeriesEEG, LFPprobeB, LFPprobeC, LFPprobeF	behavior:BehavioralTimeSeries
sub-575102_ses-20210603_behavior+ecephys	ElectricalSeriesEEG, LFPprobeB, LFPprobeC, LFPprobeF	behavior:BehavioralTimeSeries
sub-586466_ses-20210729_behavior+ecephys	ElectricalSeriesEEG, LFPprobeB, LFPprobeF	behavior:BehavioralTimeSeries
sub-586468_ses-20210819_behavior+ecephys	ElectricalSeriesEEG, LFPprobeB, LFPprobeC, LFPprobeF	behavior:BehavioralTimeSeries

File (session_id)	Acquisition	Processing modules: interfaces
sub-590479_ses-20210921_behavior+ecephys	ElectricalSeriesEEG, LFPprobeB, LFPprobeF	behavior:BehavioralTimeSeries
sub-590480_ses-20211111_behavior+ecephys	ElectricalSeriesEEG, LFPprobeB, LFPprobeC, LFPprobeF	behavior:BehavioralTimeSeries
sub-599017_ses-20220210_behavior+ecephys	ElectricalSeriesEEG, LFPprobeB, LFPprobeC, LFPprobeF	behavior:BehavioralTimeSeries

4. Units per region (Computed by this export)



5. All region labels (Computed by this export)

One row per region label exactly as stored in the NWB files, all units (no quality filter), sorted by unit count. Firing-rate columns summarise the NWB units.firing_rate values where the dataset provides them.

Region label	Units	Files	Subjects
MOs	1822	18	17
LD	1036	14	13
ACAv	814	11	10
CA1	590	17	16
SSp-tr	529	10	9
ACAd	527	16	15
VISp	452	9	8
CA3	440	14	13
VAL	427	7	6
PL	402	4	4
LSr	374	4	4
LP	304	5	5
SSp-ll	283	10	9
MGv	255	4	3
MGm	240	6	5
PO	234	3	3
VPM	194	6	5
VPL	177	5	4
ProS	175	4	4
DG-mo	171	11	10
DG-sg	164	11	10
CP	148	1	1
fp	138	11	10
AD	134	2	2
SSp-bfd	128	2	2
ZI	121	5	4
MGd	114	5	5
SGN	109	3	3
VISrl	99	2	2
or	95	11	10
alv	92	15	14
RSPagl	86	2	2
ccg	85	6	6
cing	80	14	13
CA2	77	11	11
fi	74	7	6
VISpm	73	1	1
DG-po	65	7	7
STR	60	4	4
ccb	57	14	13

Region label	Units	Files	Subjects
CL	56	1	1
LGd-ip	49	1	1
PoT	48	1	1
root	48	11	10
VISa	44	2	2
Eth	44	1	1
TH	35	5	4
IGL	32	1	1
bsc	30	5	5
RT	30	1	1
VISam	30	1	1
LGd-sh	28	3	2
PIL	27	1	1
TTd	27	1	1
AV	22	1	1
LGd-co	19	4	4
LGv	18	1	1
fiber tracts	17	5	5
scwm	15	2	2
ILA	15	1	1
IG	12	3	2
SNr	10	1	1
int	10	1	1
ar	7	1	1
PP	6	1	1
IAD	5	1	1
MD	5	1	1
MB	5	1	1
em	4	1	1
HPF	3	1	1
DP	3	1	1

6. Intervals tables (stimuli, trials, epochs)

Each NWB intervals table, with the number of files containing it, total rows, and its columns as stored.

Table	Files	Rows	Columns
epochs	24	48	start_time, stop_time
trials	24	33,841	start_time, stop_time, stimulus_type, stimulus_description, estim_current, estim_target_region, estim_target_depth, behavioral_epoch, is_running, is_valid

7. Column and signal descriptions from the NWB files

Quoted verbatim from sub-521885_ses-20200709_behavior+ecephys.nwb; all files in the Dandiset share this structure.

Location in NWB file	Name	Description (as written in the NWB file)
electrodes	(table)	metadata about extracellular electrodes
electrodes	location	the location of channel within the subject e.g. brain region
electrodes	group	a reference to the ElectrodeGroup this electrode is a part of
electrodes	group_name	the name of the ElectrodeGroup this electrode is a part of
electrodes	is_data_valid	True for electrodes/channels with usable data
electrodes	probe_vertical_position	length-wise position of electrode/channel on device (microns)
electrodes	probe_horizontal_position	width-wise position of electrode/channel on device (microns)
electrodes	x	the x coordinate of the position (+x is posterior)
electrodes	y	the y coordinate of the position (+y is inferior)
electrodes	z	the z coordinate of the position (+z is right)
electrodes	reference	Description of the reference electrode and/or reference scheme used for this electrode, e.g., "stainless steel skull screw" or "online common average referencing".
intervals/epochs	(table)	experimental epochs
intervals/epochs	start_time	Start time of epoch, in seconds
intervals/epochs	stop_time	Stop time of epoch, in seconds
intervals/epochs	tags	user-defined tags
intervals/trials	(table)	experimental trials
intervals/trials	start_time	Start time of epoch, in seconds
intervals/trials	stop_time	Stop time of epoch, in seconds
intervals/trials	stimulus_type	type of stimulus delivered
intervals/trials	stimulus_description	more specific description of stimulus type
intervals/trials	estim_current	electrical stimulation current (μ A), if applicable
intervals/trials	estim_target_region	electrical stimulation target region, if applicable
intervals/trials	estim_target_depth	electrical stimulation target depth (cortical layer), if applicable
intervals/trials	behavioral_epoch	behavioral epoch when stimulus was delivered
intervals/trials	is_running	True if mouse is running during trial, running defined as average speed > 0 cm/s between [-0.5, 0.5] s from stimulus onset
intervals/trials	is_valid	True for valid trials, trials are invalid if they have large electrical artifacts or if electrical stimulation was off target
acquisition	ElectricalSeriesEEG	voltage measured over time and associated timestamps from EEG array
processing/behavior	(module)	processed behavioral data
processing/behavior/BehavioralTimeSeries	running_speed	running speed data, computed from wheel angular velocity

8. Exported files

File	Rows	Columns
tables/sessions.csv	24	session_id, asset_path, size_gb, subject_id, species, sex, age, genotype, strain, session_start_time, session_description, institution, lab, n_units, n_electrodes, electrode_groups, region_source, regions, intervals, acquisition, processing
tables/units.parquet	12,149	session_id, unit_id, unit_row, region, location, x, y, z, isi_violations, amplitude_cutoff, presence_ratio, waveform_duration, subject_id
tables/region_summary.csv	71	region, n_units, n_sessions, n_subjects
tables/intervals_summary.csv	2	table, n_sessions, n_rows, columns
intervals/<table>.parquet	one per intervals table	session_id + the table's columns
tables/dataset_metadata.json, nwb_file_metadata.json, nwb_descriptions.csv		published metadata quoted in sections 1 and 7